Current Project Analysis Document

Project Name

Crescent Chique Designs & Co.

Project Type

Python + MySQL Console-Based Interior Design Management System

Original Development Period

Approximately 2 years ago as an academic project.

Executive Summary

The current system is a console-based application developed using Python and MySQL.

The application allows users to browse interior design packages, estimate renovation costs, select design preferences, and store customer interactions in a MySQL database.

The project successfully demonstrates basic Python programming concepts, MySQL database integration, menu-driven interfaces, and CRUD operations.

However, the application was designed as an academic project and is not suitable for real-world production use.

The objective is to transform the existing application into a modern web-based SaaS platform while preserving the core business idea and valuable domain knowledge already present in the system.

Existing Business Concept

The existing project operates as an Interior Design and Renovation Service Platform.

Customers can:

- Browse predefined interior design packages
- Explore room renovation options
- Request custom renovation packages

- Receive estimated renovation costs
- Provide contact information
- Place renovation requests

Administrators can:

- Manage customer records
- Store renovation history
- Manage employee information

Existing Features

Customer Management

Current Functionality:

- Customer information collection
- Contact details storage
- Customer history tracking

Status:

Partially reusable

Recommended Action:

Redesign database structure and workflows.

Renovation Package Selection

Current Functionality:

- 1 BHK packages
- 2 BHK packages
- 3 BHK packages
- Custom renovation requests

Status:

Highly valuable

Recommended Action:

Retain concept but move package data into database tables.

Area-Based Renovation

Current Functionality:

Customers can select:

- Kitchen
- Bedroom
- Living Room
- Dining Area
- Wardrobe
- Restroom

Status:

Highly valuable

Recommended Action:

Retain and improve.

Cost Estimation

Current Functionality:

Cost estimated using:

Area × Price Per Sq Ft

Status:

Valuable

Recommended Action:

Convert into formal quotation engine.

Design Portfolio

Current Functionality:

Predefined interior designs are displayed through locally stored images.

Status:

Valuable

Recommended Action:

Convert into web-based design gallery.

MySQL Integration

Current Functionality:

Stores:

- Customer records
- Order history
- Employee information

Status:

Reusable

Recommended Action:

Redesign schema using normalized database architecture.

Current Technical Architecture

Frontend:

Python Console Interface

Backend:

Python

Database:

MySQL

Communication:

Direct SQL Queries

Image Storage:

Local Files

Authentication:

Basic Username/Password

Major Technical Limitations

Limitation 1

Console-Based User Interface

Problem:

Requires users to interact through terminal menus.

Impact:

Not suitable for customers.

Solution:

Replace with web application.

Priority:

Critical

Limitation 2

Hardcoded Business Data

Problem:

Designs, packages, prices, and locations are hardcoded inside source code.

Impact:

Difficult to maintain and scale.

Solution:

Move all business data into database tables.

Priority:

Critical

Limitation 3

Code Duplication

Problem:

Large portions of the code repeat similar logic.

Examples:

- Package selection
- State validation
- Cost calculation
- Design display

Impact:

Difficult maintenance.

Solution:

Refactor using reusable services and database-driven design.

Priority:

Critical

Limitation 4

SQL Injection Risk

Problem:

SQL queries are created using string formatting.

Impact:

Security vulnerability.

Solution:

Use SQLAlchemy ORM and parameterized queries.

Priority:

Critical

Limitation 5

Poor Scalability

Problem:

Adding new packages requires source code modification.

Impact:

High maintenance cost.

Solution:

Admin-controlled design management.

Priority:

High

Limitation 6

Local Image Dependency

Problem:

Images are loaded from local folders.

Impact:

Deployment challenges.

Solution:

Store images using cloud storage or database references.

Priority:

High

Limitation 7

Weak Authentication

Problem:

Passwords are stored directly.

Impact:

Security risk.

Solution:

bcrypt password hashing.

Priority:

Critical

Limitation 8

No Role-Based Access

Problem:

No distinction between customer and administrator.

Impact:

Poor security and access control.

Solution:

Implement RBAC.

Priority:

Critical

Business Logic Worth Preserving

The following ideas should remain in the new system:

✓ Interior design package catalogue

✓ Room-wise renovation options

✓ Area-based pricing

✓ Renovation quotation generation

✓ Customer enquiry management

✓ Design portfolio browsing

✓ Custom renovation requests

✓ Project history tracking

Business Logic To Replace

The following implementations should be removed completely:

✗ Console menus

✗ Recursive navigation

✗ Hardcoded design data

✗ Hardcoded city lists

✗ Local image display using PIL

✗ Direct SQL query strings

✗ Plain-text password storage

✗ Repetitive code blocks

Target State (Future System)

The new platform should become:

A modern web-based Interior Design Management Platform providing:

- Customer Portal
- Admin Dashboard
- Design Gallery
- Appointment Booking
- Quotation Generation
- Project Tracking
- Document Uploads
- Secure Authentication

The platform should support real customers and be deployable to production.

Migration Strategy

Phase 1

Preserve:

- Existing business idea
- Renovation package concepts
- Pricing model

Replace:

- User interface
- Database schema
- Authentication

Phase 2

Introduce:

- Design Gallery
- Appointment Booking
- Project Tracking

Phase 3

Introduce:

- Reviews
- Analytics
- Notifications

Phase 4

Introduce:

- AI Design Recommendations
- Vendor Marketplace
- Mobile Application

Conclusion

The existing project provides strong domain knowledge and a valid business concept.

The goal is not to rewrite the project from scratch but to evolve it into a scalable, secure, maintainable, and production-ready SaaS platform while preserving the successful interior design and renovation workflows already implemented.

End of Document